BiteWalk

Release Plan & Sprint Plans

Cross-Platform Mobile App (iOS & Android)

Powered by Supabase + React Native

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Executive Summary

BiteWalk is an incentive-based walking app that rewards users with discounts at local businesses based on distance walked. The app targets college students and environmentally conscious individuals, promoting physical fitness, local business support, and reduced car emissions.

This document outlines the full release plan and detailed sprint breakdowns for building BiteWalk as a cross-platform mobile application supporting both iOS and Android from day one using React Native. The backend is powered by Supabase (PostgreSQL), chosen for its native fit with BiteWalk's relational data model — points ledgers, leaderboards, fraud detection, and business analytics. Firebase Cloud Messaging is used solely for push notifications. The plan is structured around three major releases spanning approximately 24 weeks of development.

Technology Strategy

Cross-Platform Approach

To deliver BiteWalk on both iOS and Android efficiently with a small team, the recommended approach is React Native. This enables a shared codebase for both platforms, significantly reducing development time and cost while still allowing native device access for step tracking and GPS.

Backend: Supabase (Primary) + FCM (Notifications)

Supabase provides the core backend: PostgreSQL database, authentication, edge functions, real-time subscriptions, and file storage. BiteWalk's data is fundamentally relational — users earn points, redeem them at businesses, businesses manage deals, and leaderboards aggregate rankings. PostgreSQL handles all of this natively with SQL joins, transactions, and aggregations. Firebase Cloud Messaging (FCM) is added solely for push notifications on both platforms.

Recommended Tech Stack

Layer	Technology
Frontend	React Native with TypeScript
Navigation	React Navigation
State Management	Redux Toolkit / Zustand
Backend & API	Supabase (PostgreSQL + Edge Functions + Realtime)
Database	Supabase PostgreSQL with Row-Level Security (RLS)
Authentication	Supabase Auth (email, Google, Apple Sign-In)
Step Tracking	Apple HealthKit (iOS) / Google Fit API (Android)
Maps & Location	Google Maps SDK / React Native Maps

Payments	Stripe (premium subscriptions) + Apple IAP / Google Play Billing
Push Notifications	Firebase Cloud Messaging (FCM) — notifications only
File Storage	Supabase Storage (S3-compatible, for business logos & images)
CI/CD	EAS Build (Expo) or Fastlane + GitHub Actions
Analytics	Mixpanel or PostHog (open-source alternative)
Error Monitoring	Sentry

Release Roadmap Overview

The BiteWalk development is structured into three major releases, each building upon the last. This phased approach allows the team to validate core assumptions early, incorporate user feedback, and scale features progressively.

Milestone	Key Deliverables	Target Date	Success Criteria
Release 1.0(MVP)	User auth & onboarding Step tracking & points Basic discount redemption Local business listings	Weeks 1–10	500 beta sign-ups, 3+ business partners, 70% weekly retention
Release 2.0(Growth)	Premium subscription tier Gamification & streaks Social features Enhanced business dashboard	Weeks 11–18	5% premium conversion, 25% increase in DAU, 10+ partners
Release 3.0(Scale)	Sponsored deals platform Advanced analytics Multi-city expansion tools Performance optimization	Weeks 19–24	Expansion to 2+ campuses, ad revenue live, <2s load times

Release 1.0 — Minimum Viable Product

The MVP focuses on the core value proposition validated through 42 customer interviews: rewarding users with discounts at local businesses for walking. Saving money on food was identified as the strongest motivator, so the MVP prioritizes a seamless walk-to-earn-to-redeem flow.

Sprints are 2 weeks each. Release 1.0 spans Sprints 1–5 (10 weeks).

Sprint 1: Project Setup & Supabase Auth
Duration: Weeks 1–2 **Goal:** Establish the project foundation with Supabase and deliver working login/signup flows on both platforms.

User Story	Tasks	Est. Points
As a new user, I want to create an account so I can start using BiteWalk	Initialize React Native project with TypeScript Create Supabase project and configure Auth providers (email, Google, Apple) Install @supabase/supabase-js and configure client Build signup screen (Name, Email, Password fields) Build login screen with forgot password flow via Supabase Auth Implement secure token storage with Supabase session management Set up navigation structure (auth vs. main stack) Configure CI/CD pipeline for iOS and Android builds	13
As a developer, I want a scalable database so we can support core features	Design PostgreSQL schema: users, profiles, activity, points tables Write and run Supabase database migrations Configure Row-Level Security (RLS) policies for user data isolation Create database functions for points calculations (SQL stored procedures) Set up Supabase Edge Functions project for custom API logic Configure Supabase Realtime for activity subscriptions	8
As a user, I want the app to match the BiteWalk brand so it feels polished	Implement BiteWalk design system (green palette, typography) Build reusable UI components (buttons, inputs, cards) Create splash screen and app icon for both platforms	5

Sprint 2: Step Tracking & Points Engine

Duration: Weeks 3–4
Goal: Deliver real-time step tracking and the core points accumulation system backed by PostgreSQL transactions.

User Story	Tasks	Est. Points
As a user, I want my steps tracked automatically so I earn points passively	Integrate Apple HealthKit for iOS step data Integrate Google Fit API for Android step data Build background step sync service Create step-to-points conversion engine (100 points/mile)	13

	Handle permissions and onboarding for health data access	
As a user, I want to see my walking stats so I can track my progress	Build Distance Walked screen with daily total Create weekly walking chart (S–M–T–W–T–F–S) Display current reward points balance prominently Show conversion rate info (1.0 mile = 100 points)	8
As a user, I want my points to persist reliably across sessions	Build points ledger table in PostgreSQL (credits, debits, balance, timestamps) Create Supabase database function for atomic point transactions (ACID-safe) Implement local cache with async sync queue for offline walking Subscribe to Supabase Realtime channel for points balance updates Write unit tests for points calculation and transaction logic	8

Sprint 3: Business Partnerships & Discounts

Duration: Weeks 5–6
Goal: Build the discount marketplace connecting users to local business rewards.

User Story	Tasks	Est. Points
As a user, I want to browse available discounts so I can decide where to walk	Build Discounts list screen with business cards Query businesses and deals from Supabase with PostGIS geo-filtering Display business name, logo (via Supabase Storage), deal description, and points cost Add category filters (food, drinks, retail) using Supabase query filters Integrate map view showing nearby participating businesses	8
As a user, I want to redeem my points for discounts at local businesses	Build redemption flow (select deal, confirm, generate QR code) Create Supabase Edge Function for redemption: validate balance, deduct points, issue voucher atomically Implement time-limited voucher system (30-min expiry) with PostgreSQL timestamp checks Build redemption history screen querying from ledger table	13
As a business owner, I want to list my deals on BiteWalk	Create basic business partner admin portal (web, using Supabase Auth for partner accounts)	8

	Build deal CRUD with RLS policies scoped to each business owner Upload business logos and deal images to Supabase Storage Build voucher validation endpoint (Edge Function) for POS verification Seed initial partner data for Bay Area pilot businesses	

Sprint 4: Maps, Notifications & Polish

Duration: Weeks 7–8
Goal: Add location-based features and push notifications to drive daily engagement.

User Story	Tasks	Est. Points
As a user, I want to see nearby deals on a map so I can walk toward rewards	Integrate Google Maps SDK with React Native Maps Display business pins with deal previews on tap Use PostGIS spatial queries on Supabase for “deals near me” sorting Show walking route/distance to nearest deals	8
As a user, I want notifications about deals and milestones so I stay motivated	Set up Firebase Cloud Messaging (FCM) for push notifications on both platforms Store FCM device tokens in Supabase users table Build Supabase Edge Functions to trigger notifications: nearby deal, milestone, inactivity Create notification preferences screen (stored in Supabase user profile) Implement deep linking from notifications to relevant screens	8
As a user, I want a smooth, bug-free experience on my phone	Fix platform-specific UI inconsistencies (iOS vs Android) Optimize app startup time and image loading (Supabase Storage CDN) Implement error boundaries and crash reporting (Sentry) Add loading states and skeleton screens throughout	5

Sprint 5: Beta Testing & MVP Launch

Duration: Weeks 9–10
Goal: Conduct beta testing with real users and prepare for public launch on both app stores.

User Story	Tasks	Est. Points

As the team, we want to beta test with UCSC students before public launch	Deploy TestFlight build (iOS) and internal testing track (Android) Recruit 50–100 beta testers from UCSC campus Set up feedback collection (in-app form + survey) Monitor Supabase dashboard for database performance and error rates Track analytics for crash rates, retention, and points usage	5
As a user, I want to find and download BiteWalk from the app store	Prepare App Store listing (screenshots, description, keywords) Prepare Google Play Store listing Submit for App Store review (plan for 1–2 week review) Submit for Google Play review Set up app store analytics tracking	5
As the team, we want to fix critical issues found in beta	Triage and fix P0/P1 bugs from beta feedback Optimize Supabase queries based on real-world usage patterns (add indexes) Implement basic fraud protection: step count anomaly detection via SQL queries Tighten RLS policies based on security review Confirm 3+ business partners onboarded for launch	8

Release 2.0 — Growth & Monetization

Release 2.0 introduces the premium subscription tier ($4.99/month as validated in the MVP design), gamification to boost retention, and social features. Customer interviews revealed that users want the app to feel fun rather than like a chore, and that a premium tier must clearly demonstrate value over the free experience.

Release 2.0 spans Sprints 6–9 (8 weeks).

Sprint 6: Premium Subscription Tier		
Duration: Weeks 11–12 **Goal:** Launch the premium subscription with double points, VIP discounts, and no ads.		
User Story	**Tasks**	**Est. Points**
As a user, I want to upgrade to premium for better rewards	Integrate Stripe for recurring subscription payments Implement Apple IAP and Google Play Billing Build Supabase Edge Function webhook to handle Stripe/IAP events and update user plan Add subscription_tier column to users table with RLS-aware queries Build premium upsell screen (Unlock Exclusive Rewards) Create premium badge and visual indicators throughout app	13
As a premium user, I want exclusive benefits that justify the cost	Implement 2x points multiplier via Supabase database function (checks tier before crediting) Build VIP discount tier: query premium-only deals with Supabase filters Remove ads for premium subscribers (client-side tier check) Add priority support channel for premium users	8
As the team, we want to manage subscriptions and prevent churn	Build subscription management screen (cancel, change plan) Implement grace period and renewal reminders via scheduled Edge Functions Set up subscription analytics using PostgreSQL queries (conversion, churn, LTV) Create A/B test framework for pricing experiments	5

Sprint 7: Gamification & Streaks

Duration: Weeks 13–14
Goal: Add game-like mechanics to drive daily engagement and long-term retention.

User Story	Tasks	Est. Points
As a user, I want walking streaks so I stay motivated to walk daily	Build streaks table in PostgreSQL with daily streak tracking logic Create Supabase database function to calculate and update streaks on each sync Build visual streak calendar component in the app Implement streak milestones (7-day, 30-day, 100-day) with bonus point rewards Add streak protection for premium users (1 missed day forgiven, tier-aware query)	8
As a user, I want achievements and badges so walking feels like a game	Design achievements table and badge definitions in PostgreSQL Build achievement evaluation via database triggers on points/distance milestones Build achievement showcase on user profile Implement progress bars and milestone notifications (via FCM triggers) Create weekly and monthly walking challenges	8
As a user, I want avatars and customization to make the app feel personal	Build avatar selection and customization screen Store avatar assets in Supabase Storage, preferences in user profile Create avatar progression tied to walking milestones Design seasonal/limited-edition avatar items Display avatar on profile and leaderboard	5

Sprint 8: Social Features & Leaderboards

Duration: Weeks 15–16
Goal: Enable social connection and community competition to drive organic growth.

User Story	Tasks	Est. Points
As a user, I want to compete with friends on a leaderboard	Build leaderboard SQL views: weekly points, distance, using window functions for ranking Implement friend system (add by username, contacts, QR code) with friendships table Create campus-wide and friends-only leaderboard views (filtered SQL queries) Subscribe to Supabase Realtime for live leaderboard updates Add leaderboard rewards (top 3 get bonus points weekly via scheduled Edge Function)	8

As a user, I want to share my achievements on social media	Build shareable achievement cards (auto-generated images) Integrate native share sheet (Instagram stories, Twitter, etc.) Create referral system: referrals table in PostgreSQL, bonus points on sign-up via Edge Function Track referral attribution and viral coefficient	8
As a user, I want to join walking groups or challenges with others	Build groups and group_members tables with RLS for group privacy Create group walking challenges (team vs. team) with aggregate SQL scoring Build group creation, join, and management flows Implement group progress tracking via Supabase Realtime subscriptions Add group activity feed	8

Sprint 9: Business Dashboard & Release 2.0 Launch		
Duration: Weeks 17–18 **Goal:** Empower business partners with analytics and finalize the 2.0 release.		
User Story	**Tasks**	**Est. Points**
As a business owner, I want to see how BiteWalk drives traffic to my store	Build partner analytics dashboard using SQL aggregations (redemptions, foot traffic, revenue) Create PostgreSQL views for deal performance metrics (views, saves, redemptions) Build exportable CSV reports via Supabase Edge Function Implement partner notification system for low-stock deals (FCM trigger)	8
As the team, we want a stable, polished 2.0 release	Conduct regression testing across all features on both platforms Perform load testing against Supabase (connection pooling, query performance) Review and optimize all RLS policies for security and performance Update app store listings with new feature screenshots Execute coordinated launch with marketing push	5

Release 3.0 — Scale & Expansion

Release 3.0 prepares BiteWalk for multi-city expansion beyond the Bay Area, introduces the sponsored deals revenue stream, and optimizes performance for a growing user base. This release transforms BiteWalk from a campus tool into a scalable platform.

Release 3.0 spans Sprints 10–12 (6 weeks).

Sprint 10: Sponsored Deals & Ad Platform		
Duration: Weeks 19–20 **Goal:** Launch the sponsored deals marketplace as a new revenue stream.		
User Story	**Tasks**	**Est. Points**
As a business owner, I want to promote my deals to more users	Build sponsored_deals table with placement priority, budget, and scheduling Create self-serve ad purchase flow for business partners (Stripe integration) Build Supabase Edge Function to serve sponsored deals based on budget and targeting Implement impression and click tracking in PostgreSQL (event logging table) Build billing system using Stripe metered billing for sponsored placements	13
As a user, I want sponsored deals to feel relevant, not intrusive	Design native-feeling sponsored deal cards (clearly labeled) Implement location and preference-based deal targeting via PostGIS + user preferences Add user feedback on deal relevance (thumbs up/down, stored in ratings table) Build frequency capping logic in Edge Function to prevent ad fatigue	8

Sprint 11: Multi-City Expansion & Analytics		
Duration: Weeks 21–22 **Goal:** Build the infrastructure to launch BiteWalk at new campuses and cities.		
User Story	**Tasks**	**Est. Points**
As the team, we want to launch in new cities efficiently	Add city/region schema to PostgreSQL with geo-boundaries (PostGIS polygons) Build city management interface in admin panel Create business partner self-serve onboarding flow (Supabase Auth + guided setup) Implement geo-fenced content delivery (city-specific deals via PostGIS queries)	8

	Build launch playbook and partner acquisition toolkit	
As the team, we want deep analytics to guide product decisions	Build internal analytics dashboard querying PostgreSQL directly (DAU, MAU, retention) Create SQL views for cohort analysis (by campus, sign-up date, plan type) Track full funnel: install → signup → first walk → first redemption Set up scheduled Edge Functions for automated metric anomaly alerts (via FCM/email)	8

Sprint 12: Performance, Security & 3.0 Launch		
Duration: Weeks 23–24 **Goal:** Optimize for scale, harden security, and ship the 3.0 release.		
User Story	**Tasks**	**Est. Points**
As a user, I want the app to be fast and reliable at all times	Optimize PostgreSQL queries with EXPLAIN ANALYZE; add composite indexes Configure Supabase connection pooling (PgBouncer) for concurrent users Implement CDN caching for Supabase Storage assets (business images) Target <2s cold start on both platforms; optimize bundle size	8
As the team, we want robust fraud protection and security	Build step count anomaly detection: SQL-based pattern analysis on activity table Add rate limiting on redemption Edge Functions (abuse detection) Conduct security audit: RLS policy review, API endpoint testing, penetration testing Implement data encryption at rest (Supabase default) and in transit (TLS) Review and rotate all API keys and service role keys	8
As the team, we want a successful 3.0 launch across multiple cities	Final QA pass across all features on both platforms Coordinate launch at 2–3 new campuses simultaneously Execute PR and marketing campaign for expansion announcement Set up Supabase monitoring dashboard and alerts for launch week Document runbook for on-call incident response	5

Supabase Architecture Overview

The following outlines how Supabase components map to BiteWalk's feature set, providing a clear picture of the backend architecture.

Supabase Component	BiteWalk Usage	Key Advantage
PostgreSQL Database	Users, points ledger, businesses, deals, redemptions, streaks, achievements, groups, leaderboards	ACID transactions for safe point operations; SQL joins for complex queries; PostGIS for geo features
Supabase Auth	User signup/login, business partner accounts, session management, password reset	Built-in email, Google, Apple providers; JWT tokens integrate with RLS policies
Row-Level Security	Users can only see their own points; businesses can only edit their own deals; admins see everything	Security at the database layer — impossible to bypass from the client
Edge Functions	Point redemption, voucher generation, FCM notification triggers, Stripe webhooks, fraud detection	Serverless TypeScript functions; low latency; scales automatically
Realtime	Live points balance updates, leaderboard refresh, group challenge progress	WebSocket subscriptions on PostgreSQL changes; no polling needed
Supabase Storage	Business logos, deal images, avatar assets, shareable achievement cards	S3-compatible with CDN; integrates with RLS for private/public buckets
Database Functions	Points calculation, streak evaluation, achievement triggers, leaderboard ranking	Run logic inside PostgreSQL for atomicity and speed; triggered automatically

Estimated Budget Alignment

Based on the business model canvas cost estimates, here is how the sprint plan aligns with the projected budget. Note that Supabase's free tier covers the beta phase, and the Pro plan ($25/month) is sufficient through Release 2.0, keeping infrastructure costs predictable.

Category	Canvas Estimate	Sprint Allocation
App Development	$30K–$60K	Releases 1.0–3.0
Marketing Team	$2K–$5K	Sprint 5, 9, 12
Sales Team (Partner Acquisition)	$5K–$15K	Sprint 3, 9, 11
Supabase Hosting	$0–$25/mo (free → Pro)	Free tier through beta; Pro from launch

Servers & Infrastructure (CAPEX)	$50K–$100K	Significantly reduced with Supabase managed hosting

Key Risks & Mitigations

Risk	Impact	Mitigation
Step tracking inconsistency across devices	Inaccurate points, user frustration	Extensive device testing; fallback to GPS distance
Low business partner adoption	Weak discount catalog, poor user retention	Start with 3–5 strong local partners; offer free pilot period
App Store review delays	Launch timeline slips	Submit early in Sprint 5; prepare for review feedback iterations
Step count fraud/spoofing	Financial loss through fake redemptions	SQL-based anomaly detection, daily caps, GPS correlation checks
Supabase Realtime latency for offline sync	Points not updating when user comes back online	Local queue with retry logic; conflict resolution via server timestamps
Premium conversion below 5% target	Revenue model viability at risk	Strong free tier; A/B test pricing and features; interview non-converters
Non-walkable areas reduce app value	Limited adoption outside campus core	Launch in walkable Bay Area campuses first; expand strategically

Success Metrics by Release

Metric	Release 1.0	Release 2.0	Release 3.0
Active Users (DAU)	200+	500+	2,000+
Weekly Retention	70%	75%	80%
Business Partners	3–5	10–15	30+
Premium Conversion	N/A	5%	8%
Avg. Walks per User/Week	3	4	5
Cities/Campuses	1 (UCSC)	1–2	3–5

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